

PRATHYUSHA MALLELA

(971) 895-4741 — prathyushamallela3@gmail.com — [linkedin.com/in/prathyusha-mallela](https://www.linkedin.com/in/prathyusha-mallela)
website: prathyushamallela.github.io — github: github.com/prathyushamallela

SUMMARY

- Software Engineer with almost **9 years of industry experience** building **backend, distributed, and ML-driven systems** in fintech and data-intensive environments. **Work Authorization:** OPT EAD – Authorized for immediate U.S. employment.
- Graduated with an **M.S. in Computer Science**, focusing on **Artificial Intelligence**, from Oregon State University (Corvallis, OR).
- Strong proficiency in **Java and Python**, with deep **object-oriented design** expertise and experience architecting systems from first principles.
- Hands-on experience across the **entire Machine Learning lifecycle**, including ETL, feature engineering, model training, evaluation, deployment, and monitoring.
- Built and deployed **production ML models** using **PyTorch** for fraud detection and automated decisioning.
- Experience with **event-driven and distributed systems** using **Kafka, Spark, Scala-based pipelines**, and **ZooKeeper**.
- **AWS Certified Cloud Practitioner** — [Credly Public Badge](#)
- Master's projects in AI (OSU):
 - **Causal-temporal modeling** for supply-chain systems using inference-based techniques. (using Causal Inferencing and Probabilistic Graphical Models)
 - **Hybrid rule-based NLP** with **text-to-text transformer models (T5)** for solving cryptic crosswords.
- Designed and maintained **Pentaho (Kettle) ETL pipelines** for large-scale transactional ingestion and ML feature pipelines.
- Familiar with **LangChain and LangGraph** for LLM-driven workflows and AI orchestration.
- **Code, research prototypes, and system designs** available at: github.com/prathyushamallela

EDUCATION

M.S. in Computer Science, focus in AI Oregon State University, USA *Mar 2025* — GPA: 3.67

B.E. in Electrical and Electronics Engineering, PESIT (affiliated to Visvesvaraya Technological University, Belgaum), Bangalore, India *2008 – 2012*

Post Graduate Certificate Program in Internet of Things, Birla Institute of Technology and Science (BITS), Pilani, India *2019 – 2020*

TECHNICAL SKILLS

Languages: Java (Core/EE), Python

Backend & Architecture: Spring Boot, Spring MVC, Hibernate, Drools, REST APIs, SOAP, Microservices, POJO Design, AWS

ML/AI: PyTorch, CNNs, Kohonen SOMs, Feature Engineering, Model Training, Deployment, Experimentation, LLM, LangChain, LangGraph

Distributed Systems: Apache Kafka, Apache Spark, Scala, ZooKeeper, Distributed Coordination

ETL / Data Engineering: Pentaho (Kettle), BigQuery

Databases: Oracle, IBM Db2, MSSQL, MongoDB, SOQL, JDBC
DevOps / Tooling: GitHub, Docker, Maven, JUnit

PROFESSIONAL EXPERIENCE

Software Engineer - Salesforce Developer & ML Engineer
DEKA Research & Development

Apr 2026 - Present

- Delivered **Salesforce and machine learning solutions** for the complaint handling team at **Mill-yard Advanced Medical Products**, supporting critical complaint handling processes and workflow automation.
- Developed and deployed crucial **complaint handling fixes and process improvements**, with more than **80% of developed features used in production**.
- Developed an **LLM-powered summarization solution** to generate concise summaries from complaint and investigation information, improving review efficiency for complaint handling teams.
- Created **Salesforce Flows** to automate complaint field autopopulation and reduce repetitive manual data entry.
- Implemented and configured **approval processes** to support complaint handling and internal review workflows.
- Reconfigured portions of a previously hard-coded Salesforce system using **Custom Metadata Types**, enabling configurable business logic and improving system maintainability.
- Designed and developed an **ML-based issue classification system** to classify issues reported in complaints and predict the most likely cause of each complaint.
- Developed a **Python-based automated UI testing application** for the MUTS Fleet Controller using **pywinauto** and Microsoft UI Automation (UIA).
- Designed reusable automated tests to validate **application menus, UI controls, device-specific operations, and expected system behavior**, with PASS/FAIL reporting.
- Implemented **dynamic device detection, synchronization, and configuration-driven testing** to support device-dependent workflows and maintainable regression testing.
- Debugged automation failures by analyzing **UI control discovery, application state, timing, and device-dependent behavior**, improving test reliability and troubleshooting.

Software Engineer – Backend & ML Systems
Manipal Technologies Ltd

2016 – 2020

- **AI-driven Fraud Detection Solutions for international banks**
 - Developed ubiquitous fraud classification models to detect suspicious bank transactions for multiple international banks, preventing multimillion-dollar losses.
 - Achieved **83% false-positive reduction, 80% fraud capture, and 40% reduction in manual review effort**, saving over **\$2M annually**.
 - Reduced fraud false positives using **Kohonen Self-Organizing Maps (SOMs)** integrated with **Drools rule engines**.
 - Designed an **11-layer CNN-based fraud detection model** achieving **80% fraud capture**.
 - Developed a **case-worker recommendation system** to prioritize high-risk transactions and improve investigation efficiency.
 - Applied **data augmentation** techniques to generate synthetic data of fraud rules and investigation scenarios.
 - Feature Engineering: Built **near real-time transaction scoring pipelines** using **Apache Kafka** and **Pentaho (Kettle) ETL** for feature generation.

- Designed **enterprise-grade backend services** using **Java, Spring Boot, Spring MVC, Hibernate**, integrating ML models with **rule-based decision engines**.
- Implemented a modular **distributed fraud platform** leveraging **Kafka** for streaming, **ZooKeeper** for coordination, and **Spark/Scala** for large-scale analytics.
- **Anti-Money Laundering (AML) solutions**
 - Led migration from rule-only systems to **hybrid ML-driven architectures**.
 - Designed **near real-time AML pipelines** using **Kafka, Apache Spark, and Pentaho ETL**.

Product Developer

2012 – 2016

MVR Info Systems

- Built backend services for a Smart Energy Alert System using REST APIs and relational databases.
- Developed Angular-based UI workflows for real-time energy monitoring.

Salesforce Programmer Analyst

Jul 2022 – Oct 2023

Oregon State University

- Supported and enhanced **Salesforce CRM-based university applications**, including the **Graduate Committee Representative List UI** and backend services.
- Performed backend testing and issue triage for the **Exam Scheduling Form**, ensuring data validation, workflow automation, and system reliability.
- Conducted **UAT, regression testing, and workflow validation** within Agile sprint cycles.
- Diagnosed and resolved **production issues**, maintaining application stability, data integrity, and reliable user workflows.